

Case Study: Smart Waste Management System in Urban Areas

1. Introduction

Rapid urbanization has led to a significant increase in waste generation in cities. Traditional waste management systems are inefficient, leading to overflowing bins, environmental pollution, and health hazards. This case study explores innovative solutions to improve waste collection and management using creative problem-solving techniques.

2. Problem Identification

Problem Statement:

Urban areas face inefficient waste collection due to:

- Fixed collection schedules (even when bins are empty/full)
- Lack of real-time monitoring
- Poor segregation of waste
- High operational costs

Impact:

- Environmental pollution
- Spread of diseases
- Increased municipal costs
- Poor quality of life

3. Root Cause Analysis

Using **5 Whys Technique**:

1. Why is waste not collected efficiently?
→ Because collection is done on fixed schedules
2. Why fixed schedules?
→ No real-time data on bin status
3. Why no real-time data?
→ Lack of smart monitoring systems
4. Why lack of systems?
→ High implementation cost and low awareness
5. Why low adoption?
→ Limited integration of technology in municipal services

4. Creative Solution Generation

Using **Brainstorming & SCAMPER Technique**:

Solution 1: Smart Dustbins (IoT-based)

- Sensors detect fill level
- Send alerts to collection trucks

Solution 2: AI-Based Route Optimization

- Dynamic routing for garbage trucks
- Reduces fuel and time

Solution 3: Reward-Based Waste Segregation App

- Citizens earn points for proper waste segregation
- Gamification approach

Solution 4: Community Composting System

- Local compost units in societies
- Reduces transportation of wet waste

Solution 5: Drone-Based Monitoring

- Monitor dumping areas and illegal waste disposal

5. Evaluation of Solutions

Using **Feasibility, Viability, and Desirability (FVD Framework)**:

Solution	Feasibility	Viability	Desirability	Remarks
Smart Dustbins	High	Medium	High	Requires initial investment
AI Routing	High	High	Medium	Efficient for municipalities
Reward App	Medium	High	High	Needs user engagement
Composting	High	High	Medium	Works best in communities
Drone Monitoring	Low	Medium	Low	Expensive & complex

6. Decision Matrix

Criteria: Cost, Impact, Scalability, Ease of Implementation

Solution	Cost	Impact	Scalability	Total
Smart Dustbins	3	5	5	13
AI Routing	4	5	4	13
Reward App	5	4	5	14
Composting	5	4	3	12
Drone Monitoring	2	3	2	7

Best Solution: Reward-Based Waste Segregation App + Smart Dustbins (Hybrid Approach)

7. Proposed Solution

Integrated Smart Waste Management System:

- IoT-enabled smart bins
- AI-based route optimization
- Mobile app for citizen participation
- Incentives for waste segregation

8. Implementation Plan

Phase 1: Pilot in selected area

Phase 2: Deploy smart bins & app

Phase 3: Train municipal staff

Phase 4: Scale across city

9. Expected Outcomes

- 30–40% reduction in waste collection cost
- Cleaner urban environment
- Increased citizen participation
- Efficient resource utilization

10. Conclusion

This case study demonstrates how combining technology and creative thinking can solve real-world urban problems. By evaluating multiple solutions using structured frameworks, the most effective and scalable approach can be identified and implemented successfully.